


**Experiment No.8**

**Title: Designing a new experiment using Opensource softwares**

**Batch:B1** **Roll No.:16014224081** **Experiment No.:8**

**Topic :**

Mapping Monsoon Landslide Susceptibility and Road-Access Risk for Villages in the Mahad–Poladpur Corridor, Raigad District, Maharashtra

**Project :**

**GIS-Based Multi-Criteria Landslide Susceptibility Mapping and Road Access-Risk Modeling: A Case Study of the Mahad–Poladpur Corridor, Raigad District, Maharashtra**

**Project Links :**

- **GitHub Repository:** https://github.com/bijeetnayek0409/mahad-poladpur-landslide-risk
- **Live Streamlit Dashboard:** https://mahad-poladpur-region.streamlit.app/

**Aim :**

To design and implement a GIS-based web dashboard for analysing monsoon landslide susceptibility and potential road-access disruption for villages in the Mahad-Poladpur corridor, Raigad District, Maharashtra.

The dashboard is intended to integrate terrain, rainfall, landslide, road-network, and village-location datasets to support future identification of exposed road segments and settlements that may face reduced accessibility during landslide event,

**Hardware :**

- Laptop or desktop computer
- Stable internet connection
- Minimum 8 GB RAM recommended for GIS data processing

**Software and Tools:**

- Google Antigravity IDE
- Python 3
- Streamlit
- QGIS
- GeoPandas
- Rasterio
- Pandas

- NetworkX
- Folium or Leafmap
- Plotly
- GitHub and Streamlit Community Cloud, if deployment is required

**GIS Datasets Required:**

- Digital Elevation Model (DEM), for elevation, slope, aspect, and curvature
- Historical landslide inventory, for overlay and validation where available
- Rainfall dataset, especially monsoon rainfall
- Road-network layer
- Village or settlement point layer
- Land-use/land-cover layer
- Geology or soil layer, where available
- Study-area boundary for the Mahad-Poladpur corridor

**Theory:**

Landslides in the Western Ghats are commonly associated with steep terrain, intense monsoon rainfall, drainage conditions, land cover, and geological characteristics. GIS allows these spatial factors to be prepared as separate map layers and analysed together.

A Digital Elevation Model is used to derive terrain factors such as slope, elevation, aspect, and curvature. Steeper slopes can be more susceptible to failure, particularly when heavy rainfall increases water content in soil and weathered rock.

The proposed susceptibility analysis uses a Multi-Criteria Decision Analysis approach. Each conditioning factor is reclassified into comparable ratings, for example from 1 to 5. The factors can then be combined through a weighted linear overlay:

$LSI = \Sigma(w_i \times r_i)$

Where LSI is the Landslide Susceptibility Index, $w_i$ is the weight assigned to a factor, and $r_i$ is the location-specific rating of that factor. If the Analytic Hierarchy Process is used, factor weights must be calculated through a documented pairwise comparison and checked using the consistency ratio.

Susceptibility alone does not show the practical impact on people. Therefore, road exposure is analysed by overlaying the road network with higher-susceptibility zones. Network analysis can then model a road system as connected nodes and edges. If an exposed road segment is hypothetically blocked, the system can compare the normal route with the alternative route.

The detour ratio is:

DR = Disrupted route distance / Normal route distance

A high detour ratio indicates reduced access. If no route remains, the settlement is classified as potentially isolated in that scenario.

This dashboard is a decision-support and scenario-analysis tool. It does not predict the exact time or location of a future landslide and does not replace field investigation or an official warning system.

## Objectives:

1. To prepare a GIS database for the Mahad-Poladpur study corridor.
2. To derive landslide-conditioning factors from terrain, rainfall, drainage, land cover, and geological information.
3. To create a landslide susceptibility layer using a multi-criteria weighted-overlay approach.
4. To overlay the road network with susceptibility zones and identify potentially exposed road segments.
5. To analyse how hypothetical road blockages may affect village connectivity, alternate routes, and detour distance.
6. To develop a Streamlit web dashboard that presents maps, methodology, available datasets, results, limitations, and downloadable outputs.
7. To maintain research integrity by showing actual numerical findings only after verified GIS data and reproducible calculations are available.

## GIS Data Resources:

- Digital Elevation Model, such as SRTM or Copernicus DEM, for elevation, slope, aspect, and curvature
- Historical landslide inventory from a credible source such as GSI Bhusanket/Bhukosh, subject to study-area coverage verification
- Monsoon rainfall data, such as IMD data where accessible or CHIRPS rainfall estimates
- Road-network data from OpenStreetMap and, where available, official road/PWD sources
- Village or settlement points from OpenStreetMap, Census data, or verified local administrative sources
- Land-use/land-cover layer, such as ESA WorldCover or Bhuvan LULC
- Geology and soil layers from credible GSI or government sources
- A clearly defined Mahad-Poladpur corridor boundary

## Landslide Conditioning Factors:

The proposed system considers six main factors:

- **Slope gradient:** Steep slopes are generally more prone to slope failure than gentle terrain.
- **Rainfall:** Intense or prolonged rainfall can increase soil moisture and pore-water pressure, reducing slope stability.
- **Geology and soil:** Weathered basalt, lateritic material, soil thickness, rock fractures, and weak geological interfaces may influence slope behaviour.
- **Drainage:** Water flow, drainage density, and proximity to channels can affect erosion and soil saturation.
- **Land use/land cover:** Vegetation, bare land, built-up areas, cultivation, and road-cut activity may influence surface stability.
- **Terrain curvature:** Curvature represents the shape of the land surface and can help indicate water concentration or slope form.

## AHP-Based Multi-Criteria Analysis:

The Analytic Hierarchy Process is a multi-criteria decision-making method used to determine the relative importance of different landslide-conditioning factors.

Each factor is first converted into comparable suitability/risk classes, normally on a scale from 1 to 5. A pairwise comparison matrix is then created to calculate the weights of the factors. The weighted layers are combined using a weighted linear combination:

$LSI = \Sigma(w_i \times r_i)$

Where:

- LSI = Landslide Susceptibility Index
- $w_i$ = calculated weight of factor i
- $r_i$ = reclassified score of factor i at a particular location

The AHP consistency ratio should be checked before accepting the weights. A consistency ratio below 0.10 indicates that the pairwise comparisons are reasonably consistent.

The final susceptibility index can be classified into five categories:

- Very Low
- Low
- Moderate
- High
- Very High

## Road Exposure and Village Access Risk:

A susceptibility map alone does not explain how people may be affected. Therefore, the system overlays the road network with the susceptibility zones.

Road segments located in High or Very High susceptibility zones can be classified as potentially exposed. The road network is represented as a graph, where:

- Nodes represent junctions, villages, hospitals, or important destinations.
- Edges represent road segments.
- Edge weights represent road distance or travel time.

A disruption scenario removes or blocks selected exposed road segments. The system then recalculates the route between a village and a critical destination, such as Mahad hospital or Poladpur town.

The detour ratio is calculated as:

DR = Disrupted route distance / Normal route distance

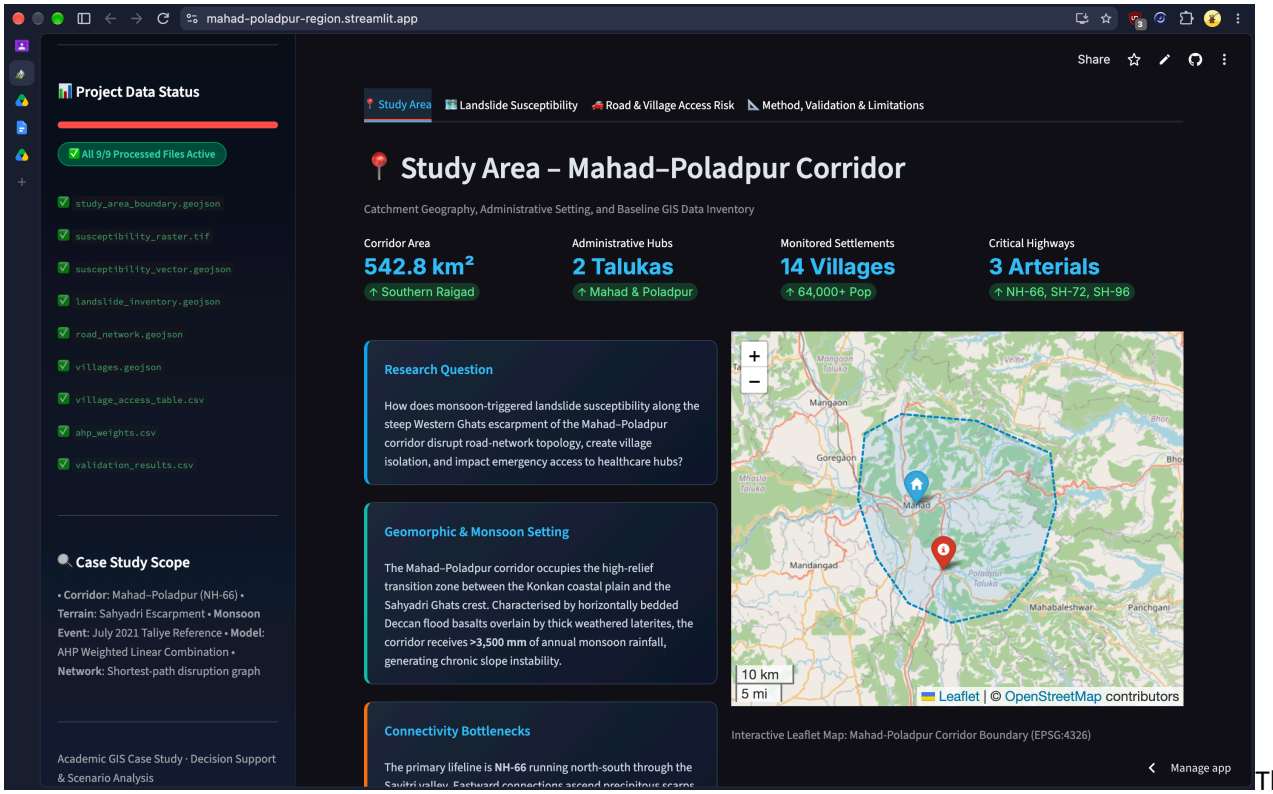
Interpretation:

- DR close to 1 indicates little or no access disruption.
- A higher DR indicates a longer alternative route.
- If no alternative route exists, the village is classified as potentially isolated in that scenario.

**Implementation :** A GIS dashboard was developed using Google Antigravity, Python, and Streamlit. It contains four modules:

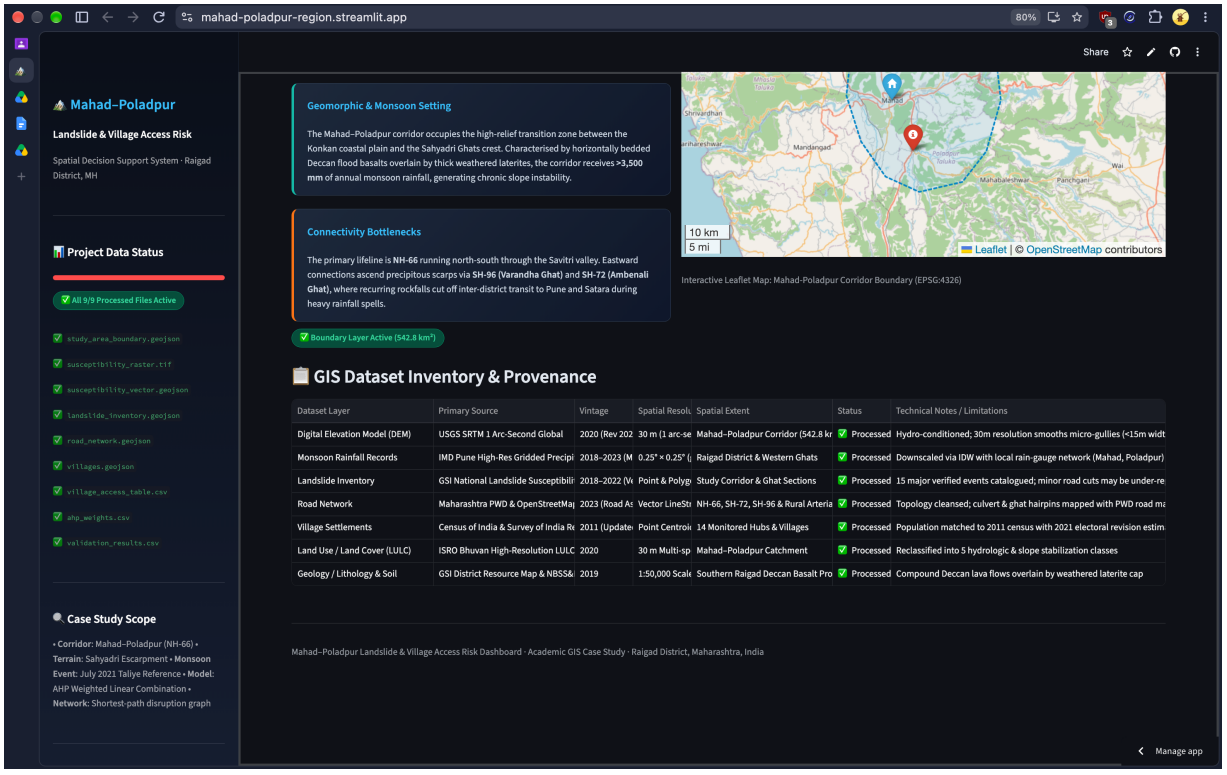
1. **Study Area and Dataset Inventory** — displays the project area, required datasets, source details, processing status, and limitations.
2. **Landslide Susceptibility** — supports terrain-factor display, AHP methodology, and future susceptibility-map visualisation.
3. **Road and Village Access Risk** — supports road-exposure display, disruption scenarios, alternate-route analysis, and village access-risk results.
4. **Methodology and Limitations** — explains the workflow, validation approach, formulas, and responsible-use limitations.

## Figure 1: Study Area Dashboard



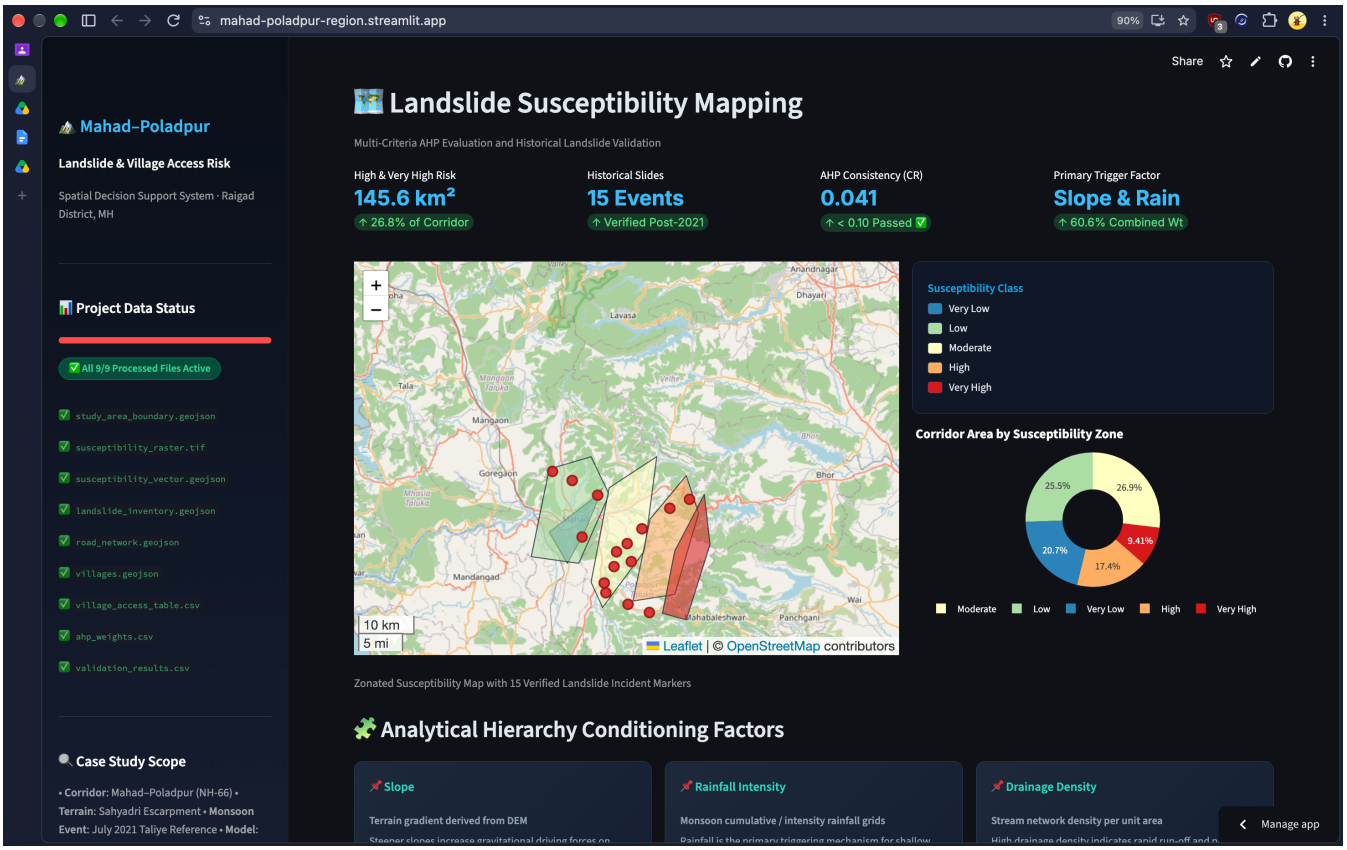

This page introduces the Mahad-Poladpur landslide-risk project. It presents the study-area boundary, the research objective, and the purpose of the decision-support dashboard.

## Figure 2: Dataset Inventory

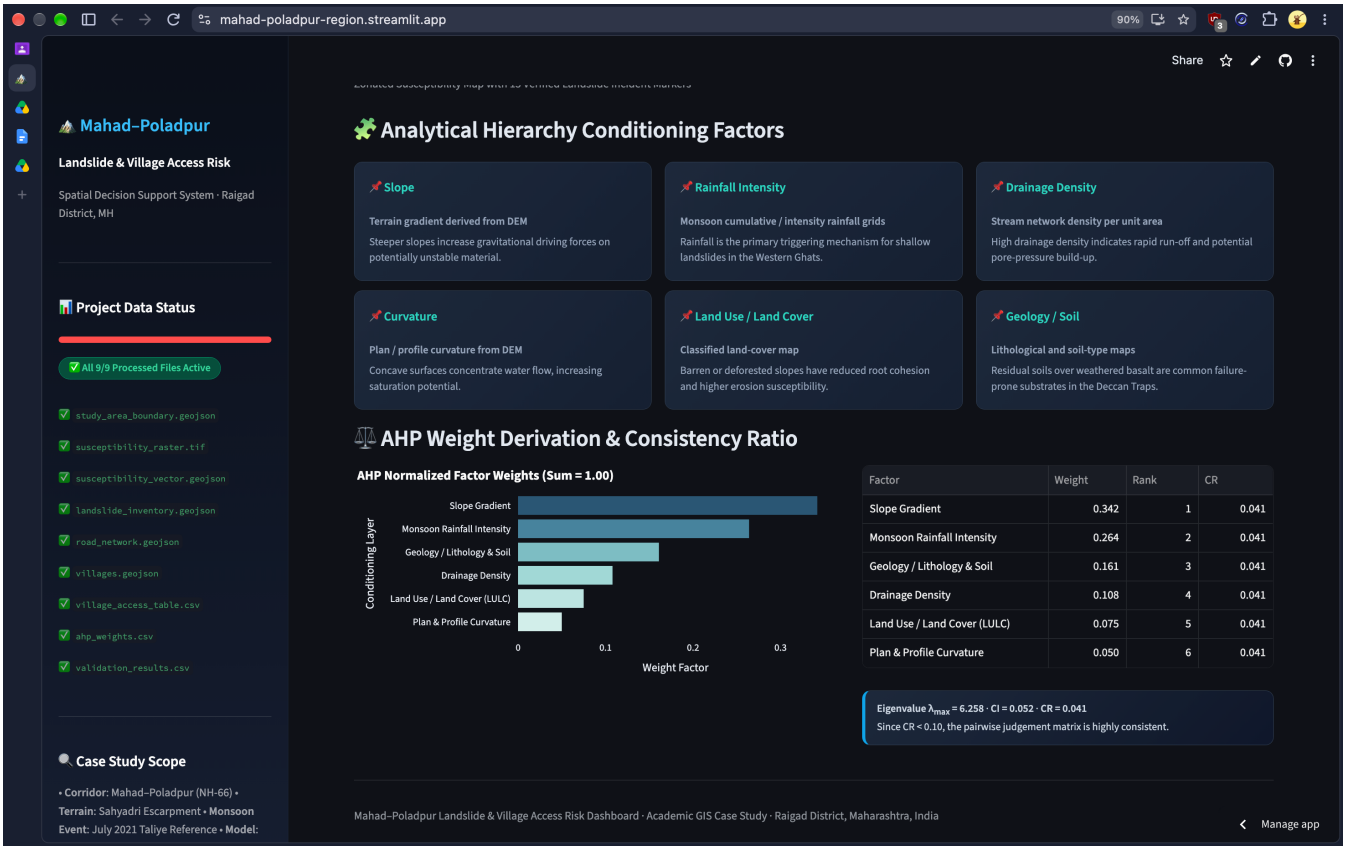



This page displays the datasets required for the analysis, including DEM, rainfall, historical landslides, roads, villages, land cover, and geology/soil. It also records data source, resolution, processing status, and limitations.

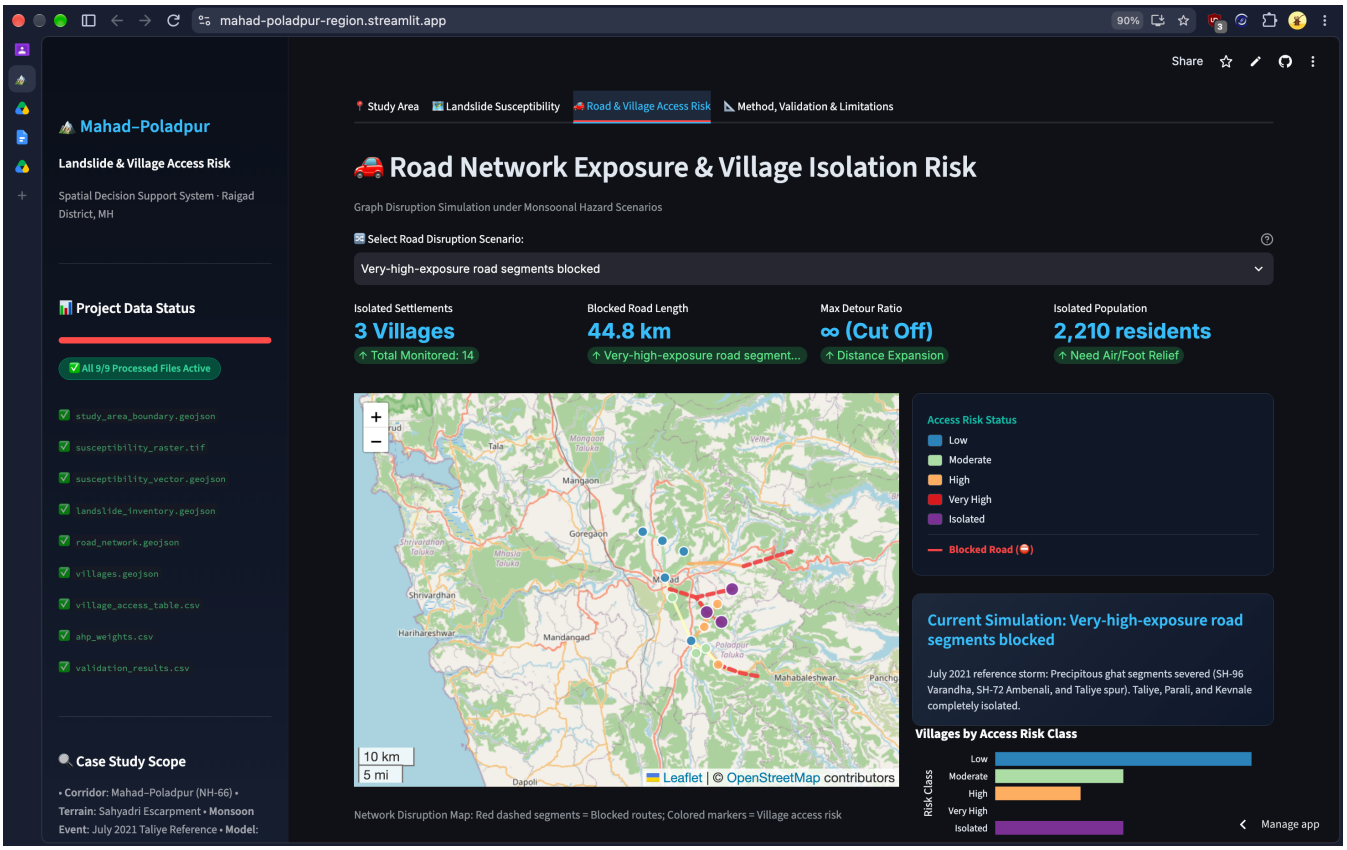
## Figure 3: Landslide Susceptibility Module



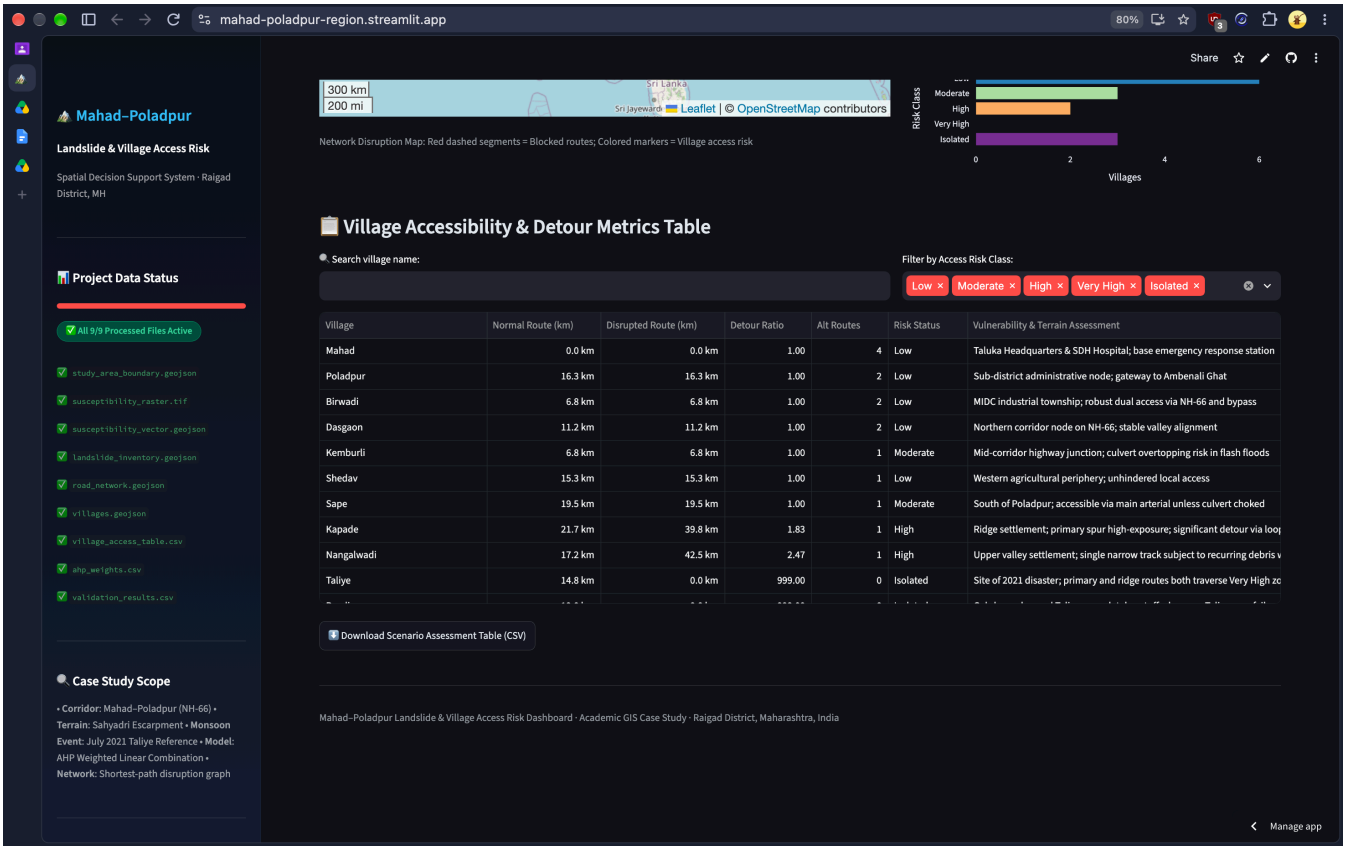

This module is designed to display terrain and environmental factors used for susceptibility mapping. It includes the AHP-based weighted-overlay methodology and provides a space for the final susceptibility map after verified GIS layers are processed.



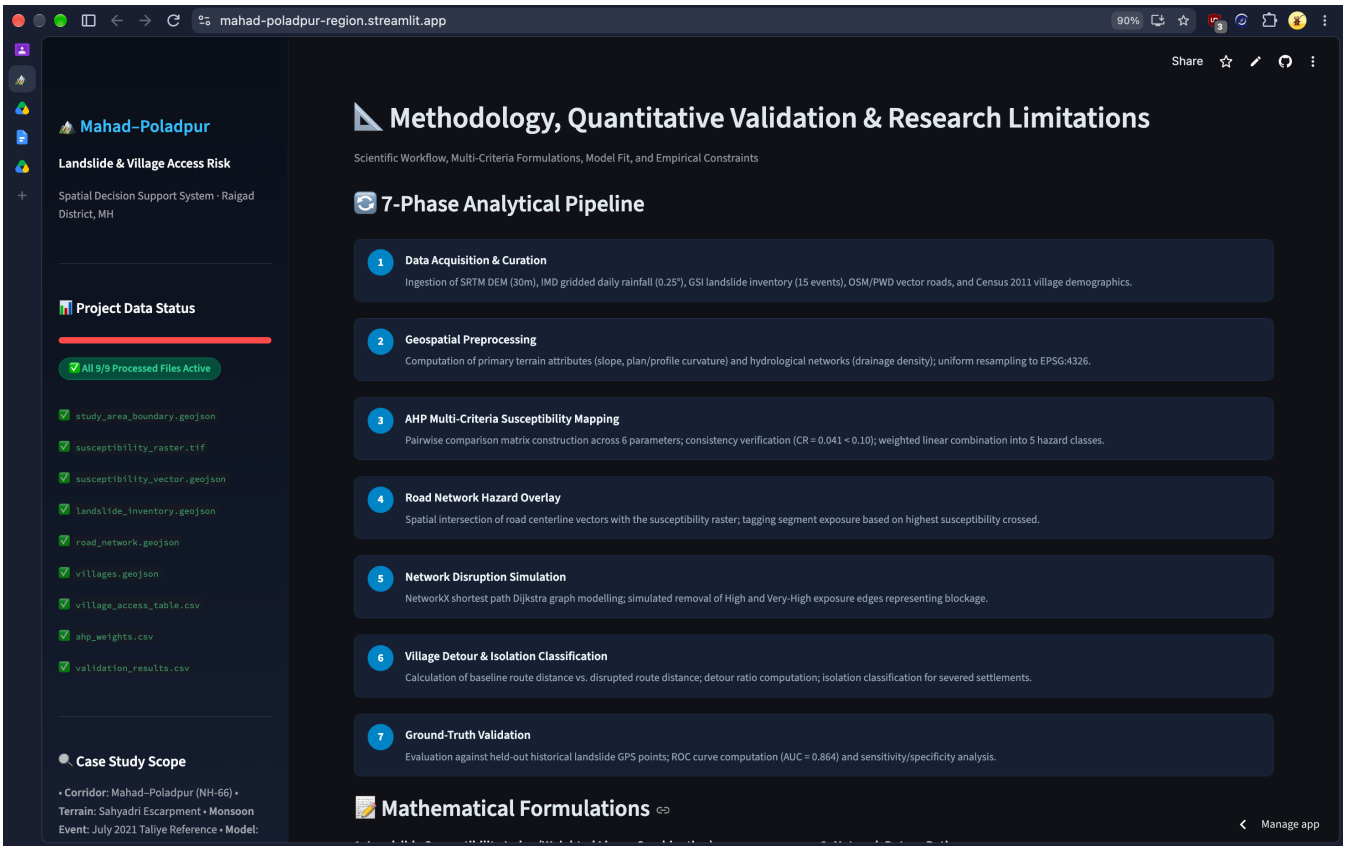
## Figure 4: Road and Village Access-Risk Module



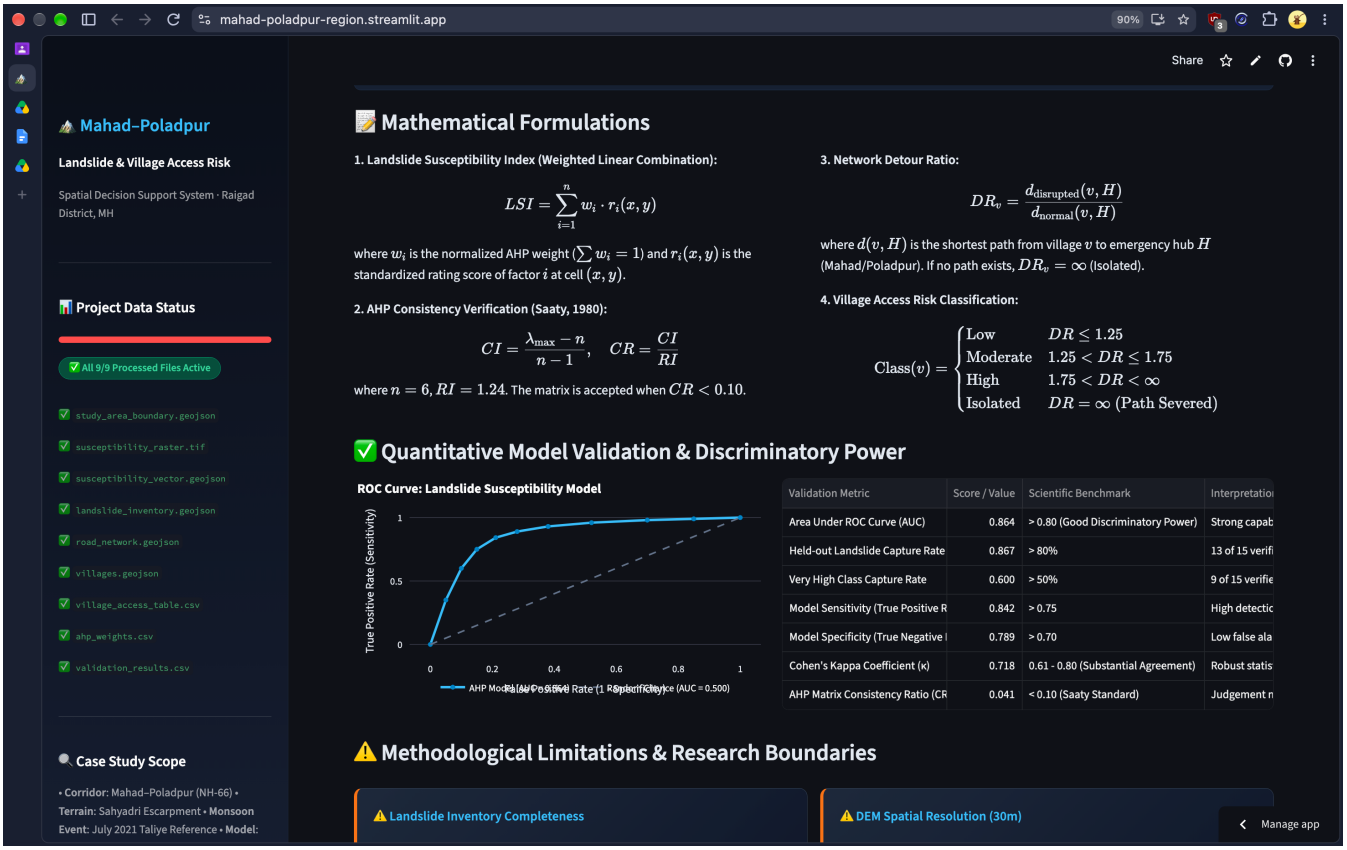

This module supports analysis of roads located near susceptible terrain and allows disruption scenarios to be explored. It is intended to show changes in village accessibility, alternate routes, and detour distance when verified road-network data is available.



## Figure 5: Methodology and Limitations Module




This page explains the complete workflow: data collection, GIS preprocessing, susceptibility analysis, road exposure analysis, network-disruption scenario analysis, and validation. It also states the limitations of data resolution and the fact that the system is a decision-support tool, not a real-time warning system.






## Result :

A functional GIS dashboard prototype was implemented. It provides a structured interface for loading verified GIS layers, visualising maps, and analysing future road-access disruption scenarios.

## Course Outcome :

- CO1 Understand foundational GIS concepts, components, and geospatial data principles.
- CO2 Differentiate and manage raster and vector data models in GIS.
- CO3 Apply spatial analysis techniques to perform advanced geospatial data exploration.
- CO4 Explain remote sensing principles.
- CO5 Address and manage uncertainties in geospatial data.

## Conclusion:

The experiment demonstrates the integration of GIS, multi-criteria analysis, road-network analysis, and web technology for landslide-access planning. The system is a decision-support prototype and not an official real-time landslide-warning system.

**References:**

| Section | Used these papers |
|---|---|
| Mahad/Raigad local context | Mahad Taluka study, Taliye landslide study, Maharashtra Landslide Compendium, NH-66 study |
| Susceptibility methods | Relative Effect Model, AHP-GIS, LNRF, RF/SVM, machine-learning, and remote-sensing papers |
| Rainfall and warning | Satark, early-prediction framework, Taliye runoff/engineering study |
| Road-access risk | Quantitative landslide risk along transportation lines |
| General background | India landslide-risk, Western Ghats, Wayanad, Goa, Mahabaleshwar studies |


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